1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 10 | 12 | 0 | 10 | 0 | 11

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Miguel Angel Virto UPF ID: U 2 7 3 9 5 1

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   $\sqrt{4} = 2$

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-2
1-8
12'5 / 7 / 9 / 7 / 2.5 / 7 / 5 / 7

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Jana Buendia UPF ID: U 2 3 3 9 1 9

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

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   (b) unbiased and inconsistent.
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   (d) biased and inconsistent.

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   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

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   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

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   (b) 1.96.
   (c) 4.96.
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   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

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   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 9 . 10 . 11 . 12 . 13 . 14 . 15

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
17'5 | 10 | 8 | 0 | 9 | 9 | 0 | 7

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Roger Torrents UPF ID: U 2 7 3 9 4 8

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

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   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
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   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

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   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

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7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
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   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 , 9 , 10 / 11 / 12 / . . . .

1-8 9 , 10 / 11 , 12 / . . . .

1-8 9 10 11 12 13 14 15
17 5 / 7 / 10 / 0 / 12 / 8 / 0 / 9

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Alex Mayné UPF ID: U213904

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
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3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
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   (c) 4.96.
   (d) .04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

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   (c) is unchanged.
   (d) is divided by two.

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1-8 9 10 11 12 13 14 15
12 | 8 | 10 | 11 | 12 | 12 | 11 | 11
1-8 9 b 11 12 13 14 15
20 | 6 | 7.5 | 10 | 0 | 5.5 | 0 | 10

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Lorenzo Mario Carolyn UPF ID: U M279935

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

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   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
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4. A Type I error occurs when a test:

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5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

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1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
17'5 | 8 | 10 | 11 | 12 | 13 | 11 | 15

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 8 | 8 | 12 | 9 | 6 | 0 | 12

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Víctor Bonet Ollé UPF ID: U 2 7 3 9 1 8

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1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 7 | 0 | 4 | 10 | 5 | 2

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Mathieu Cohen UPF ID: U285023

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

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   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
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   (c) rejects a false null hypothesis.
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   (b) 1.96.
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   (c) is unchanged.
   (d) is divided by two.

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   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
17'5 | 8 | 10 | 0 | 0 | 7 | 0 | 12

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Tom Masson Bigas UPF ID: U273941

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \dots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent. b) ✓
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence. ✗

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$. ✗
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$. ✓
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ. ✗

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $t = \frac{3 - \mu_0}{1} = 1{,}96$ $\quad -\mu_0 = 1{,}96 - 3 \quad \mu_0 = 1{,}04.$

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y / \sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean: $\sqrt{1} = 1 \quad \sqrt{4} = 2$

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
15 | 3 | 9 | 0 | 12 | 7 | 0 | 12

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Cristian Jiménez García UPF ID: U273912

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 10 | 12 | 1 | 7 | 2 | 8.5

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Aros Fornas UPF ID: U273911

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent. ~ var doesn't change
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance. –
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$. = $\frac{\bar{Y}-\mu}{\sigma_{\bar{Y}}}$ $\sigma_{\bar{Y}} = \frac{\sigma_Y}{\sqrt{n}}$
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis. –
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

   $\frac{3-0}{1} = 3$ $1'96 = \frac{3-\mu_0}{1}$ $1'04 = \mu_0$

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

   $\sqrt{100} = 10$ $\sqrt{400} = 20$

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 6/8 | 6 | 9 | 0 | 0 | 5

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Blanca Navas Cerezuela UPF ID: U 2 7 3 9 4 3

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 7 | 10 | 12 | 0 | 4 | 0 | 9

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Mark Bjorca UPF ID: U273936

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\tilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

I CHOOSE (c) !!!

(a) unbiased and consistent.
(b) unbiased and inconsistent.
(c) biased and consistent.
(d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

(a) Zero covariance implies independence.
(b) Zero covariance implies $E[Y \mid X] = E[Y]$.
(c) Independence implies zero covariance.
(d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

(a) $\bar{Y}$ has exactly a normal distribution for every n.
(b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
(c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
(d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

(a) rejects a true null hypothesis.
(b) does not reject a false null hypothesis.
(c) rejects a false null hypothesis.
(d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

(a) 0.
(b) 1.96.
(c) 4.96.
(d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

(a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
(b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
(c) $U_i = \widehat{U}_i$ for every observation.
(d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

(a) is divided by four.
(b) is multiplied by two.
(c) is unchanged.
(d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

(a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
(b) $SER = RMSE$ for every finite sample.
(c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
(d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 9 | 10 | 6 | 12 | 4 | 0 | 10

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Roger Tello Tarrés UPF ID: U273931

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $t_{obs} = \frac{\bar{Y} - \mu_0}{SE(\bar{Y})}$

   To not reject, $|t| \geq 1.96$.

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y / \sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 10 | 12 | 12 | 0 | 0 | 2

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Alba Cañas Moreno UPF ID: U 273938

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\tilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

consistent $\tilde{\mu}_n \xrightarrow{p} \mu$ as $n \to \infty$

→ (a) unbiased and consistent. → (b) unbiased and inconsistent.
(c) biased and consistent. (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

(a) Zero covariance implies independence. (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
(c) Independence implies zero covariance. (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

→ (a) $\bar{Y}$ has exactly a normal distribution for every n. → (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
(c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$. (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

$\frac{\bar{y} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$

4. A Type I error occurs when a test:

→ (a) rejects a true null hypothesis. (b) does not reject a false null hypothesis. - Type 2
(c) rejects a false null hypothesis. (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

$t: \frac{3 - \mu_Y = 0}{1} = 3 > 1.96$

(a) 0. (b) 1.96.
(c) 4.96. (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

(a) $\sum_i \hat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$. → (b) $\sum_i \hat{U}_i = 0$ and $\sum_i X_i \hat{U}_i = 0$, even if the regressor is endogenous.
(c) $U_i = \hat{U}_i$ for every observation. (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

(a) is divided by four. (b) is multiplied by two.
(c) is unchanged. (d) is divided by two. → $\frac{\sigma_Y}{\sqrt{4n}} = \frac{\sigma_Y}{2\sqrt{n}}$

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

(a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$. (b) $SER = RMSE$ for every finite sample.
(c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$. (d) Both equal $\sqrt{SSR/(n-1)}$.

$SER = \sqrt{\frac{SSR}{n-2}}$

$RMSE = \sqrt{\frac{SSR}{n}}$

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
7.5 | 8 | 10 | 0 | 12 | 10 | 0 | 12

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Margal Bundó UPF ID: U226580

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

   CI = $3 \pm 1 \cdot 1.96 = [1.04, 1.96]$

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 5.5 | 8 | 6 | 2 | 8 | 0 | 0

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Roger Riba Fernández UPF ID: U273947

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent. ✗
   (d) biased and inconsistent. ✗

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

   $t_{obs} = \frac{x-3}{SE}$ $1.96 = |x-3|$ $|x| = 1.04$

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 7.5 | 10 | 0 | 12 | 8.5 | 1 | 1

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Alijandra González Lanumbe UPF ID: U 2 7 3 9 2 1

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ. LLN

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean: $\frac{0.4}{\sqrt{n}}$

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 / 9 / 10 / 11 / 12 / 13 / 14 / 15
12'5 / 4.5 / 6 / 0 / 0 / 9 / 0 / 5

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Carlos Bolaños Ferrer UPF ID: U273907

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
15 | 7.5 | 10 | 6 | 6 | 0 | 0 | 0

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

Name and surname: Alex Vladomiru Gispert UPF ID: U227565

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected? $t = \frac{|\bar{Y} - \mu_0|}{SE} = 3 > 1.96 \Rightarrow \mu_0 = 1.96\, SE(\bar{Y})$

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)? $SER = \frac{SSR}{n-2}$

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8	9	10	11	12	13	14	15
17'5	10	10	1	12	12	5	2

Econometrics (25117): Midterm 1, 2026–2027

Duration: 180 minutes

Monday, 9 November 2026

41603718F

Name and surname: Berta Badosa Regi UPF ID: U _ _ _ _ _ _ _

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 8 | 9 | 6 | 10.5 | 7 | 4 | 9

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: JOSEP RUSSO RARÍS UPF ID: U273926

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\tilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

   $E(\tilde{\mu}_n) = \mu$
   $\tilde{\mu}_n \xrightarrow{p} \mu$

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \hat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \hat{U}_i = 0$ and $\sum_i X_i \hat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \hat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

   $\frac{\sigma_Y}{\sqrt{100}} = \frac{\sigma_Y}{10}$; $\frac{\sigma_Y}{20}$

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

8	9	10	11	12	13	14	15
20	9	8	~~7~~	1	9	5	3

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Jan Garcia Vargalleo UPF ID: U 273920

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $\mathrm{E}[Y \mid X] = \mathrm{E}[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $\mathrm{E}[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
17.5 | 5 | 8 | 12 | 3 | 6 | 4 | 0

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Maria Pineda UPF ID: U209850

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
   (d) Perfect positive correlation implies independence.

3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

1-8 | 9 | 10 | 11 | 12 | 13 | 14 | 15
20 | 10 | 10 | 19 | 12 | 9 | 0 | 0

Econometrics (25117): Midterm 1, 2026–2027
Duration: 180 minutes
Monday, 9 November 2026

Name and surname: Elisabet Vaca UPF ID: U2 73949

Answer all questions. Questions 1–8 have one correct option each: 2.5 points for a correct answer, 0 for a wrong or blank answer. Questions 9–15 are worth 80 points; show calculations and briefly explain your reasoning on separate answer paper. The total is 100 points; the grade out of 10 is the total divided by 10. For inference, use the large-sample standard normal approximation. The two-sided critical values are 1.96 at 5% and 2.576 at 1%. Exact fractions, radicals or sensible rounding are acceptable. All numerical examples are hypothetical.

Multiple-choice questions

1. Let $Y_1, \ldots, Y_n$ be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Consider $\widetilde{\mu}_n = Y_1$, which uses only the first observation, regardless of n. This estimator is:

   (a) unbiased and consistent.
   (b) unbiased and inconsistent.
   (c) biased and consistent.
   (d) biased and inconsistent.

2. Which statement always holds when the required moments exist?

   (a) Zero covariance implies independence.
   (b) Zero covariance implies $E[Y \mid X] = E[Y]$.
   (c) Independence implies zero covariance.
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3. For i.i.d. observations with mean μ and variance $0 < \sigma^2 < \infty$, the central limit theorem states that, as $n \to \infty$:

   (a) $\bar{Y}$ has exactly a normal distribution for every n.
   (b) $\sqrt{n}(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (c) $(\bar{Y} - \mu)/\sigma \xrightarrow{d} N(0,1)$.
   (d) $\bar{Y} \xrightarrow{p} 0$, whatever the value of μ.

4. A Type I error occurs when a test:

   (a) rejects a true null hypothesis.
   (b) does not reject a false null hypothesis.
   (c) rejects a false null hypothesis.
   (d) does not reject a true null hypothesis.

5. A large-sample two-sided test uses an estimate $\bar{y} = 3$ and $SE(\bar{Y}) = 1$. The null $H_0 : \mu_Y = 0$ is rejected at 5%. For any hypothesized value μ_0, reject when $|t| > 1.96$. What is the smallest μ_0 for which $H_0 : \mu_Y = \mu_0$ would not be rejected?

   $t = \frac{3 - \mu_0}{\wedge} = 1.96 \longrightarrow \mu_0 = 1.04$

   (a) 0.
   (b) 1.96.
   (c) 4.96.
   (d) 1.04.

6. In an OLS regression with an intercept and a nonconstant regressor, which statement is necessarily true, apart from numerical rounding?

   (a) $\sum_i \widehat{U}_i = 0$ proves that $E[U_i \mid X_i] = 0$.
   (b) $\sum_i \widehat{U}_i = 0$ and $\sum_i X_i \widehat{U}_i = 0$, even if the regressor is endogenous.
   (c) $U_i = \widehat{U}_i$ for every observation.
   (d) Every individual residual equals zero.

7. For i.i.d. observations with a finite, positive population variance, the standard deviation of the sample mean is $\sigma_Y/\sqrt{n}$. If the sample size increases from 100 to 400 while the population distribution stays the same, the standard deviation of the sample mean:

   (a) is divided by four.
   (b) is multiplied by two.
   (c) is unchanged.
   (d) is divided by two.

8. For simple OLS with an intercept, $n > 2$ and $SSR > 0$, which statement correctly distinguishes the standard error of the regression (SER) from the root mean squared residual ($RMSE$)?

   (a) $SER = \sqrt{SSR/n}$; $RMSE = \sqrt{SSR/(n-2)}$.
   (b) $SER = RMSE$ for every finite sample.
   (c) $SER = \sqrt{SSR/(n-2)}$; $RMSE = \sqrt{SSR/n}$.
   (d) Both equal $\sqrt{SSR/(n-1)}$.

NOT 0.6 independent.